Question 4 — Sentiment Classification and Text-Generation Evaluation

Given data

The LLM classifies 300 reviews into Positive, Negative, and Neutral. Confusion-matrix rows are actual labels; columns are predicted labels.

Stage 1 — Zero-shot baseline

Actual / Predicted	Positive	Negative	Neutral	Row total
Positive	100	5	15	120
Negative	8	80	12	100
Neutral	20	10	50	80
Column total	**128**	**95**	**77**	**300**

Stage 2 — Improved prompt

The revised prompt defines Neutral as “factual, mixed, or no clear opinion” and includes six few-shot examples, two per class, including borderline mixed reviews.

Actual / Predicted	Positive	Negative	Neutral	Row total
Positive	106	4	10	120
Negative	6	84	10	100
Neutral	9	7	64	80
Column total	**121**	**95**	**84**	**300**

1. Stage 1 overall accuracy with a 95% confidence interval

Accuracy is the number of correct predictions divided by the total number of reviews. The correct predictions are the diagonal entries of the confusion matrix.

$$\text{Correct} = 100 + 80 + 50 = 230$$

$$\text{Accuracy} = \frac{230}{300} = 0.7666667$$

$$\boxed{\text{Stage 1 accuracy} = 76.67\%}$$

95% confidence interval — Wilson score method

Use the Wilson score interval, with $n = 300$, $\hat{p} = 230/300 = 0.7666667$, and $z = 1.96$ for a 95% confidence level.

$$\text{Denominator} = 1 + \frac{z^2}{n}$$

$$\text{Center} = \frac{\hat{p} + \frac{z^2}{2n}}{1 + \frac{z^2}{n}}$$

$$\text{Half-width} = \frac{z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}$$

Substituting the values gives:

$$\text{Center} = 0.7632951, \qquad \text{Half-width} = 0.0476774$$

Therefore:

$$CI_{95\%} = 0.7632951 \pm 0.0476774$$

$$\boxed{CI_{95\%} = [0.71562,\ 0.81097]}$$

Stage 1 result: Accuracy is **76.67%**, with a 95% Wilson confidence interval of **71.56% to 81.10%**.

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2. Stage 1 precision, recall, and F1 for each class

For each class:

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{TP}{\text{predicted column total}}$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{TP}{\text{actual row total}}$$

$$F1 = \frac{2(\text{Precision})(\text{Recall})}{\text{Precision} + \text{Recall}}$$

Equivalently:

$$F1 = \frac{2TP}{2TP + FP + FN}$$

Positive

True positives = 100. The predicted-Positive column total is 128 and the actual-Positive row total is 120.

$$P_{+} = \frac{100}{128} = 0.78125$$

$$R_{+} = \frac{100}{120} = 0.8333333$$

$$F1_{+} = \frac{2(100)}{2(100) + (128 - 100) + (120 - 100)}$$

$$F1_{+} = \frac{200}{248} = 0.8064516$$

Negative

True positives = 80. The predicted-Negative column total is 95 and the actual-Negative row total is 100.

$$P_{-} = \frac{80}{95} = 0.8421053$$

$$R_{-} = \frac{80}{100} = 0.8000000$$

$$F1_{-} = \frac{2(80)}{2(80) + (95 - 80) + (100 - 80)}$$

$$F1_{-} = \frac{160}{195} = 0.8205128$$

Neutral

True positives = 50. The predicted-Neutral column total is 77 and the actual-Neutral row total is 80.

$$P_N = \frac{50}{77} = 0.6493506$$

$$R_N = \frac{50}{80} = 0.6250000$$

$$F1_N = \frac{2(50)}{2(50) + (77 - 50) + (80 - 50)}$$

$$F1_N = \frac{100}{157} = 0.6369427$$

Stage 1 class-wise results

Class	Precision	Recall	F1
Positive	0.7813	0.8333	0.8065
Negative	0.8421	0.8000	0.8205
Neutral	0.6494	0.6250	0.6369

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3. Stage 1 macro, weighted, and micro F1

Macro F1

Macro F1 gives equal weight to all three classes:

$$F1_{\text{macro}} = \frac{F1_{+} + F1_{-} + F1_N}{3}$$

$$= \frac{0.8064516 + 0.8205128 + 0.6369427}{3}$$

$$\boxed{F1_{\text{macro}} = 0.7546357}$$

Weighted F1

Weighted F1 weights each class by its actual support:

- Positive support = 120
- Negative support = 100
- Neutral support = 80

$$F1_{\text{weighted}} = \frac{120F1_{+} + 100F1_{-} + 80F1_{N}}{300}$$

$$= \frac{120(0.8064516) + 100(0.8205128) + 80(0.6369427)}{300}$$

$$\boxed{F1_{\text{weighted}} = 0.7659363}$$

Micro F1

For single-label multiclass classification, micro-averaged precision, recall, and F1 are equal to overall accuracy.

$$F1_{\text{micro}} = \frac{\sum TP}{300}$$

$$= \frac{100 + 80 + 50}{300}$$

$$\boxed{F1_{\text{micro}} = 0.7666667}$$

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4. Interpret Stage 1: where does the LLM fail, and what should change?

The main weakness is **Neutral sentiment**.

- Neutral precision is 0.6494 (64.94%).
- Neutral recall is 0.6250 (62.50%).
- Neutral F1 is 0.6369 (63.69%), the lowest of the three classes.
- Of the 80 actually Neutral reviews, 20 were incorrectly predicted as Positive and 10 as Negative. Therefore, 30 of 80 Neutral reviews were misclassified.

The largest individual confusion is **actual Neutral predicted as Positive**: 20 reviews. A further 15 actual Positive reviews were predicted as Neutral. This suggests the zero-shot prompt does not provide a sufficiently clear boundary between Neutral and Positive, especially for mixed or factual reviews.

Recommended prompt changes

1. Define Neutral explicitly as factual, mixed, or having no clear opinion.
2. Add few-shot examples for all three classes, especially borderline mixed reviews.
3. Include examples where a review contains both positive and negative statements, so the model does not classify using only one sentiment-bearing phrase.
4. Tell the model to assess the overall opinion of the review rather than treating every positive or negative word as decisive.
5. Keep the class definitions and output constraints consistent across examples.

These recommendations directly target the observed Neutral/Positive confusion and the weak Neutral recall.

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5. Stage 2: re-evaluate and compare with Stage 1

Use the same formulas and the same 300 reviews.

5.1 Stage 2 accuracy and 95% confidence interval

Correct predictions are the diagonal entries:

$$\text{Correct} = 106 + 84 + 64 = 254$$

$$\text{Accuracy} = \frac{254}{300} = 0.8466667$$

$$\boxed{\text{Stage 2 accuracy} = 84.67\%}$$

For the Wilson interval, use $\hat{p} = 254/300 = 0.8466667$, $n = 300$, and $z = 1.96$.

$$\text{Center} = 0.8422836, \qquad \text{Half-width} = 0.0407506$$

$$CI_{95\%} = 0.8422836 \pm 0.0407506$$

$$\boxed{CI_{95\%} = [0.80153,\ 0.88303]}$$

Stage 2 result: Accuracy is **84.67%**, with a 95% Wilson confidence interval of **80.15% to 88.30%**.

5.2 Stage 2 precision, recall, and F1

Positive

$$P_+ = \frac{106}{121} = 0.8760331$$

$$R_+ = \frac{106}{120} = 0.8833333$$

$$F1_+ = \frac{2(106)}{2(106) + (121 - 106) + (120 - 106)}$$

$$F1_+ = \frac{212}{241} = 0.8796680$$

Negative

$$P_- = \frac{84}{95} = 0.8842105$$

$$R_- = \frac{84}{100} = 0.8400000$$

$$F1_- = \frac{2(84)}{2(84) + (95 - 84) + (100 - 84)}$$

$$F1_- = \frac{168}{195} = 0.8615385$$

Neutral

$$P_N = \frac{64}{84} = 0.7619048$$

$$R_N = \frac{64}{80} = 0.8000000$$

$$F1_N = \frac{2(64)}{2(64) + (84 - 64) + (80 - 64)}$$

$$F1_N = \frac{128}{164} = 0.7804878$$

Stage 2 class-wise results

Class	Precision	Recall	F1
Positive	0.8760	0.8833	0.8797
Negative	0.8842	0.8400	0.8615
Neutral	0.7619	0.8000	0.7805

5.3 Stage 2 macro, weighted, and micro F1

Macro F1:

$$F1_{\text{macro}} = \frac{0.8796680 + 0.8615385 + 0.7804878}{3}$$

$$\boxed{F1_{\text{macro}} = 0.8405648}$$

Weighted F1:

$$F1_{\text{weighted}} = \frac{120(0.8796680) + 100(0.8615385) + 80(0.7804878)}{300}$$

$$\boxed{F1_{\text{weighted}} = 0.8471768}$$

Micro F1:

$$F1_{\text{micro}} = \frac{106 + 84 + 64}{300}$$

$$\boxed{F1_{\text{micro}} = 0.8466667}$$

5.4 Stage 1 versus Stage 2

Metric	Stage 1	Stage 2	Change
Accuracy	0.7667	0.8467	+0.0800
Macro F1	0.7546	0.8406	+0.0859

Metric	Stage 1	Stage 2	Change
Weighted F1	0.7659	0.8472	+0.0812
Micro F1	0.7667	0.8467	+0.0800

Accuracy increases from 76.67% to 84.67%, an improvement of **8.00 percentage points**. The improved prompt classifies 254 rather than 230 reviews correctly: **24 additional correct predictions**.

All three classes improve in F1:

Class	Stage 1 F1	Stage 2 F1	Improvement
Positive	0.8065	0.8797	+0.0732
Negative	0.8205	0.8615	+0.0410
Neutral	0.6369	0.7805	+0.1435

Neutral has the largest F1 improvement. Its recall rises from 62.50% to 80.00%, showing that the explicit definition and borderline few-shot examples address much of the original weakness.

Some errors remain: 10 actual Positive reviews and 10 actual Negative reviews are still classified as Neutral; 9 actual Neutral reviews are classified as Positive and 7 as Negative. The revised prompt improves the classification substantially, but borderline sentiment remains the main source of confusion.

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6. Few-shot prompt stability: mean and standard deviation

The five accuracies from different example orderings are:

$$0.847,\ 0.840,\ 0.853,\ 0.837,\ 0.850$$

Mean

$$\bar{x} = \frac{\sum_{i=1}^{5} x_i}{5}$$

$$\bar{x} = \frac{0.847 + 0.840 + 0.853 + 0.837 + 0.850}{5}$$

$$\boxed{\bar{x} = 0.8454 = 84.54\%}$$

Sample standard deviation

Use the sample standard deviation with denominator $n-1$:

$$s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}$$

For these five runs:

$$\boxed{s = 0.0067305}$$

Therefore:

$$\boxed{\text{Mean accuracy} \pm \text{SD} = 0.8454 \pm 0.00673}$$

Equivalently:

$$\boxed{84.54\% \pm 0.673 \text{ percentage points}}$$

The observed range is 83.7% to 85.3%, a spread of 1.6 percentage points. The standard deviation is relatively small compared with the mean, so the prompt appears reasonably stable across these five example orderings. However, ordering still causes measurable variation; the result is not completely invariant to example order.

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7. Evaluate the generated justification using BLEU-4, ROUGE-1, ROUGE-2, ROUGE-L, and METEOR

The assignment supplies normalized text: lowercase and punctuation removed.

Reference:

> the delivery was late but the product quality is good

Candidate:

> the product quality is good but delivery was slow

Token counts:

- Reference: 10 tokens.

- Candidate: 9 tokens.

For reproducibility, use whitespace-separated tokens and exact token matches for BLEU and ROUGE. Report ROUGE F1 scores together with their precision and recall. For METEOR, use exact unigram matching only (no stemming or synonym expansion) and the common parameters $\alpha = 0.9$, $\beta = 3$, and $\gamma = 0.5$.

7.1 BLEU-4

BLEU-4 uses modified n-gram precision for 1-grams through 4-grams, their geometric mean, and a brevity penalty.

$$BLEU\text{-}4 = BP \times \exp\left(\frac{1}{4}\sum_{n=1}^{4} \ln p_n\right)$$

The modified n-gram precisions are:

N-gram	Candidate n-grams	Matching n-grams	Precision
1-gram	9	8	$8/9 = 0.888889$
2-gram	8	5	$5/8 = 0.625000$
3-gram	7	3	$3/7 = 0.428571$
4-gram	6	2	$2/6 = 0.333333$

The candidate is shorter than the reference, so the brevity penalty is:

$$BP = \exp\left(1 - \frac{r}{c}\right)$$

where reference length $r = 10$ and candidate length $c = 9$.

$$BP = \exp\left(1 - \frac{10}{9}\right)$$

$$BP = 0.8948393$$

Substitute the four n-gram precisions:

$$BLEU\text{-}4 = 0.8948393 \times \exp\left[\frac{\ln(8/9) + \ln(5/8) + \ln(3/7) + \ln(2/6)}{4}\right]$$

$$\boxed{BLEU\text{-}4 = 0.474955 \approx 0.4750}$$

7.2 ROUGE-1

ROUGE-1 measures unigram overlap.

There are 8 matching unigrams, 9 candidate tokens, and 10 reference tokens.

$$P = \frac{8}{9} = 0.8888889$$

$$R = \frac{8}{10} = 0.8000000$$

$$F1 = \frac{2PR}{P + R}$$

$$\boxed{ROUGE\text{-}1\ F1 = 0.842105 \approx 0.8421}$$

7.3 ROUGE-2

ROUGE-2 measures bigram overlap.

There are 5 matching bigrams, 8 candidate bigrams, and 9 reference bigrams.

$$P = \frac{5}{8} = 0.6250000$$

$$R = \frac{5}{9} = 0.5555556$$

$$F1 = \frac{2(0.6250000)(0.5555556)}{0.6250000 + 0.5555556}$$

$$\boxed{ROUGE\text{-}2\ F1 = 0.588235 \approx 0.5882}$$

7.4 ROUGE-L

ROUGE-L uses the longest common subsequence (LCS), which preserves word order but does not require matched words to be adjacent.

The longest common subsequence is:

> the product quality is good

Its length is 5 tokens.

$$P_{LCS} = \frac{5}{9} = 0.5555556$$

$$R_{LCS} = \frac{5}{10} = 0.5000000$$

$$F1_{LCS} = \frac{2 P_{LCS} R_{LCS}}{P_{LCS} + R_{LCS}}$$

$$\boxed{ROUGE\text{-}L\ F1 = 0.526316 \approx 0.5263}$$

7.5 METEOR

Using exact unigram matching:

- Matching unigrams $m = 8$.
- Candidate length $t = 9$.
- Reference length $r = 10$.
- Number of alignment chunks $ch = 3$.

Unigram precision:

$$P = \frac{m}{t} = \frac{8}{9} = 0.8888889$$

Unigram recall:

$$R = \frac{m}{r} = \frac{8}{10} = 0.8000000$$

METEOR's recall-weighted harmonic mean, using $\alpha = 0.9$, is:

$$F_{\text{mean}} = \frac{PR}{0.9P + 0.1R}$$

Equivalently:

$$F_{\text{mean}} = \frac{10PR}{R + 9P}$$

Substitute:

$$F_{\text{mean}} = \frac{10(0.8888889)(0.8)}{0.8 + 9(0.8888889)}$$

$$F_{\text{mean}} = 0.8080808$$

The fragmentation penalty is:

$$Penalty = \gamma \left(\frac{ch}{m} \right)^{\beta}$$

Using $\gamma = 0.5$, $\beta = 3$, $ch = 3$, and $m = 8$:

$$Penalty = 0.5 \left(\frac{3}{8} \right)^{3}$$

$$Penalty = 0.0263672$$

The METEOR score is:

$$METEOR = F_{\text{mean}}(1 - Penalty)$$

$$METEOR = 0.8080808(1 - 0.0263672)$$

$$\boxed{METEOR = 0.786774 \approx 0.7868}$$

Text metric results

Metric	Score
BLEU-4	**0.4750**
ROUGE-1 F1	**0.8421**
ROUGE-2 F1	**0.5882**
ROUGE-L F1	**0.5263**
METEOR	**0.7868**

The candidate preserves most of the reference's individual words, which supports its relatively high ROUGE-1 and METEOR scores. However, the wording order changes: the candidate moves “the product quality is good” before “but delivery was slow.” This reduces bigram, longer n-gram, and longest-common-subsequence overlap. “Late” is also replaced with “slow,” so exact n-gram metrics do not count that word as a match. These metrics

measure lexical overlap and ordering; they do not fully measure semantic equivalence.

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8. Perplexity of the generated justification

The assignment gives the probabilities for the 9 generated tokens:

$$[0.60,\ 0.40,\ 0.50,\ 0.80,\ 0.70,\ 0.30,\ 0.25,\ 0.50,\ 0.20]$$

Perplexity is the exponential of the average negative log probability:

$$PPL = \exp\left(-\frac{1}{N}\sum_{i=1}^{N}\ln p_i\right)$$

Here, $N = 9$.

Token position	Probability	Natural log
1	0.60	-0.510826
2	0.40	-0.916291
3	0.50	-0.693147
4	0.80	-0.223144
5	0.70	-0.356675
6	0.30	-1.203973
7	0.25	-1.386294
8	0.50	-0.693147
9	0.20	-1.609438
Total	—	**-7.592934**

Sum of log probabilities:

$$\sum_{i=1}^{9}\ln p_i = -7.592934289893$$

Average negative log probability:

$$-\frac{1}{9}(-7.592934289893) = 0.843659365544$$

Therefore:

$$PPL = \exp(0.843659365544)$$

$$\boxed{PPL = 2.324859 \approx 2.325}$$

Interpretation: The model's geometric-mean probability for the generated tokens is approximately $1/2.325$, or 0.430. Perplexity reflects how much probability the model assigns to the observed generated tokens: lower perplexity means higher average token probability under that model. It does not, by itself, establish that the justification is factually correct, logically sound, or semantically faithful to the reference.